

Solar System, stars, orbits and satellites

Space physics
Paper 2 · Separate Physics only
26 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 12 starred (★) questions first.

	QUESTION	ANSWER	✓
1	Name the main objects in our Solar System.	The Sun, eight planets, dwarf planets, moons, asteroids and comets	☐
2	Name the eight planets in order from the Sun.	Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune	☐
3	What is a galaxy?	A huge collection of stars held together by gravity.	☐
4	Give the name of our galaxy.	The Milky Way	☐
5	Name a cloud of dust and gas where stars form.	A nebula	☐
6 ★	Name the force that pulls a nebula together to form a star.	Gravity	☐
7	What is a protostar?	A hot ball of gas formed as gravity pulls a nebula together, before fusion starts.	☐
8 ★	Name the process that begins when a protostar becomes hot enough.	Nuclear fusion	☐
9 ★	Explain why a main sequence star is stable.	The inward pull of gravity is balanced by the outward push from fusion energy.	☐
10	State what a Sun-sized star becomes after its main sequence stage.	A red giant, then a white dwarf, then a black dwarf	☐
11	State what a star much bigger than the Sun becomes after its main sequence stage.	A red super giant, then a supernova, then a neutron star or a black hole	☐
12 ★	State how elements up to iron are formed.	By nuclear fusion in stars.	☐

	QUESTION	ANSWER	✓
13 ★	State how elements heavier than iron are formed.	In supernovae	☐
14 ★	State what keeps one object in orbit around another.	Gravity	☐
15	What is an orbit?	The curved path of one object around another in space.	☐
16	Name a natural satellite of Earth.	The Moon	☐
17	What is an artificial satellite?	A human-made object in orbit around a planet.	☐
18 ★	Explain why an object in orbit is accelerating.	Its direction, and so its velocity, is always changing.	☐
19 ★	Explain why the velocity of an object in a circular orbit changes but its speed does not.	Gravity acts at right angles to its motion, changing its direction but not its speed.	☐
20 ★	State how the speed of a stable orbit depends on its radius.	The smaller the radius, the greater the speed.	☐
21	State what the Sun was formed from.	A nebula (a cloud of dust and gas) pulled together by gravity.	☐
22	Name the natural satellites of planets.	Moons	☐
23 ★	List the stages in the life cycle of a Sun-sized star.	Nebula, protostar, main sequence star, red giant, white dwarf, black dwarf	☐
24 ★	List the stages in the life cycle of a star much bigger than the Sun.	Nebula, protostar, main sequence star, red super giant, supernova, then a neutron star or a black hole	☐
25 ★	State how elements made in stars are spread through the Universe.	By supernova explosions.	☐
26	What do planets orbit? What do moons and artificial satellites orbit?	Planets orbit the Sun. Moons and artificial satellites orbit planets.	☐